Log Insights

Every number has a receipt.

Engineer, team lead and manager ask the same logs. SereneDB computes the evidence, Claude explains it, and each role leaves with its own decision.

4,747,945

BGL log events, full span r_8f3bdd12.f4

207,820

OpenStack log events, full span r_2c4302ad.f4

8 tools

one registry for MCP and web chat

Every figure links to its original query receipt

WHY NOT JUST CHAT WITH YOUR LOGS?

Chat over logs guesses. Log Insights measures.

TYPICAL CHAT OVER LOGS		LOG INSIGHTS
Retrieves 50 lines and guesses how often an error happens.	→	**Frequencies come from SQL over the whole scope.** Search shows examples, aggregation counts, truncation is always stated.
The model adds, divides and ranks numbers itself.	→	**Every number cites a fact ID.** `check_claims` verifies value, metric and group against the stored receipt.
The manager gets a shorter engineer answer.	→	**Roles are playbooks.** They change the question, the evidence, the check and the next step.
A co-occurrence is sold as the root cause.	→	**Observation, hypothesis and cause stay apart.** Each hypothesis names the check that would decide it.

SERENEDB DOES THE WORK

From raw line to ranked priority, inside the database

1 Parse

`read_csv` + `regexp_extract` split raw lines in SQL. Rejects keep their reason: 18 r_45c7a4ec.f59 BGL lines.

→

2 Normalize

21 SQL pattern macros turn messages into templates. Diagnostic codes such as `status: 503` stay.

→

3 Index

ICU analyzer, inverted index, BM25 ranking and phrase positions from the index.

→

4 Analyze

`ts_phrase`, `ts_levenshtein`, `MATCH_RECOGNIZE` episodes, `quantile_cont`, window-function rankings.

WHAT CLAUDE RECEIVES

Counts, shares, deltas, episodes, percentiles, rankings and rule outcomes, computed by SereneDB. Never raw rows to add up.

ALSO NATIVE SERENEDB

Remote Parquet read in place over HTTPS, the built-in OTLP/HTTP listener, vector and hybrid search with RRF.

No parsing scripts and no client-side math. TypeScript validates inputs, assembles allowlisted SQL, assigns fact IDs and stores receipts. Function-by-function receipts: submission/serenedb-functions.md

Four queries changed the next step

Question: Control streams fail during job start. Is one node broken, or the network?

search_logs — 19 ms

5,983

exact matches for the phrase "failed to read message prefix"

Hits truncated, count exact

r_b63b6c6e.f4

→

recurring_patterns — 39 ms

7 episodes

on 512 nodes in 32 racks. **Not one node.**

r_bcd0408f.f8 r_bcd0408f.f12

r_bcd0408f.f13

→

get_log_context — 16 ms

-838.005 ms

before the first failure, on another node:

```
ciod: LOGIN chdir(/p/gb2/draeger/
benchmark/dat32k_060205) failed:
No such file or directory
```

r_b6ce5f6c.f8

→

context aggregate — 44 ms

6 of 7

episode starts follow that chdir error within 10 min. Background: 23 of 3,896 windows.

Ratio 145.193 r_16d9e58b.f9

Recorded before the check

Systemic job/network event at the I/O nodes

one node: contradicted | cause: open

Next step after the check

Check working directory and program image at job start before any network repair.

temporal association: supported, not proven

ROLE-APPROPRIATE INSIGHT

Same case. Three roles. Three different decisions.

ENGINEER

Is one node at fault?

Symptoms on 512 r_bcd0408f.f12 nodes. A missing working directory precedes 6 of 7 episode starts r_16d9e58b.f6 . Initiating component: open.

DECISION

Match job-start directory and filesystem availability to the failing streams.

TEAM LEAD

What keeps recurring?

Program-image load errors: 22 r_8207d0e9.f16 episodes on 15 r_8207d0e9.f13 active days. Prior month is outside the BGL span: no growth claimed.

DECISION

Workload Runtime reviews program-image and directory validation at startup.

MANAGER

What gets effort first?

Line volume points to control-stream errors: 5,983 r_8207d0e9.f11 events. Episodes point to program-image errors: 22 r_8207d0e9.f9 versus 7 r_8207d0e9.f38 .

DECISION

Fund a bounded job-start check, not the loudest pattern. Business impact: unknown.

Analyst-written views from the real tool calls on slide 4. SereneDB ranks all 8 candidates by episodes, then breadth; no risk score. Ownership is fictional.

THE AGENT IN ACTION

Claude answers live through MCP, with citations

ENGINEER / OPENSTACK

Claude Desktop / Opus 5.5 / guidance 1.0.8

2 matching log events [r_0bb58204.f3]

"The identity endpoint returned an error page instead of token data. Status: open. This rests on log wording only."

NEXT STEP

Then check the identity service (Keystone) logs and health at both timestamps.

TEAM LEAD / BGL

Claude Code / Opus 5.5 / guidance 1.0.7

22 episodes [r_07df2980.f8]

"Workload Runtime should do a root-cause analysis of the program-image load failures in `ciod`:"

NEXT STEP

Check whether the episodes line up with particular job submissions or paths. Treat any such link as an association, not a confirmed cause.

MANAGER / BGL

Claude Code / Opus 5.5 / guidance 1.0.7

289 episodes [r_d1c334b8.f14]

"Assign Compute Platform (fictional owner of KERNEL) to investigate the leading pattern, since it has the most episodes and is widespread."

NEXT STEP

Business impact unknown

Original answer excerpts, unedited, from Claude Desktop and Claude Code with Opus 5.5. Complete answers, tool traces and receipts: submission/agent-examples/

SHOW THE WORK

Don't trust the model. Open the receipt.

Receipt	**r_16d9e58b / get_log_context 1.0.0**
Agent input	`mode: aggregate, window_minutes: 10, pattern_id: 287312f13c189c4d, dataset: bgl, from: 2005-06-01T00:00:00Z, to: 2005-07-01T00:00:00Z, direction: before, top_n: 5`
Scope	5,983 log events / truncated: `false`
Data version	`dv_229e422f86a9fc6e`
Result digest	`f2b31164b07510028fbf23caa22a25d9…`
Database	44 ms over 3 statements

EXECUTED SQL, STATEMENT 3 OF 3 (EXCERPT)

```
WITH matched AS (SELECT log_events.event_id, log_events.ts, log_events.pattern_id FROM
log_events WHERE log_events.dataset = $1 AND log_events.ts >= $2::TIMESTAMPTZ AND
log_events.ts < $3::TIMESTAMPTZ AND log_events.pattern_id = $4), anchors AS (SELECT
event_id, ts FROM matched MATCH_RECOGNIZE (ORDER BY ts, event_id MEASURES
FIRST(event_id) AS event_id, FIRST(ts) AS ts ONE ROW PER MATCH AFTER MATCH SKIP PAST
LAST ROW PATTERN (s n*) DEFINE n AS ts - PREV(ts) < $5::INTERVAL)), excluded AS (SELECT
DISTINCT pattern_id FROM matched WHERE pattern_id IS NOT NULL), hits AS (SELECT 'before'
AS direction, e.pattern_id, count(DISTINCT a.event_id) AS anchors_with_pattern,
min(e.event_id) AS example_event_id FROM anchors a JOIN log_events e ON e.dataset = $6
AND e.ts >= a.ts - $7::INTERVAL AND e.ts < a.ts WHERE e.pattern_id NOT IN (SELECT
pattern_id FROM excluded) GROUP BY e.pattern_id) …
```

CHECK_CLAIMS CATCHES A WRONG NUMBER

Claim: 90,406 request IDs — **VERIFIED**

Claim: 90,407 request IDs — **MISMATCH**

Fact r_097256b2.f1, request IDs over the full OpenStack span

WORKED CASE

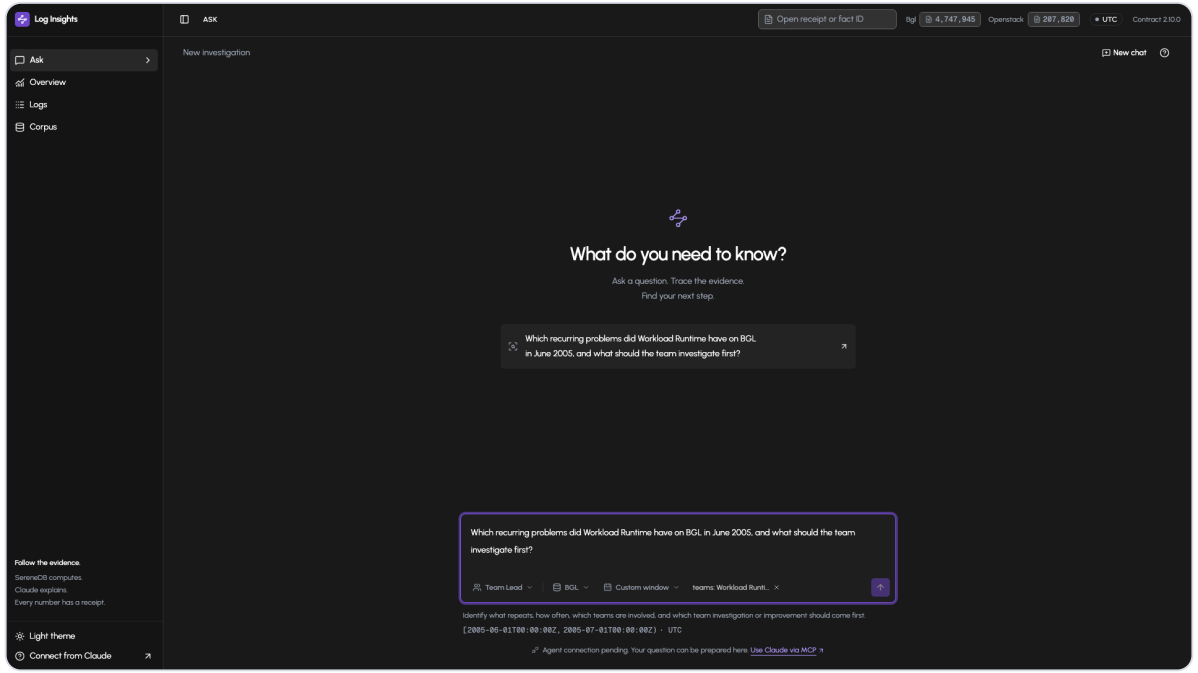
Cited values verified in three batches: 30, 30, 8.
Mismatches: 0, 0, 0. Unresolved: 0, 0, 0.

RE-RUN

3 of 3 registry re-runs return the same signature, data version and result digest.

The agent sends validated parameters, never SQL. A re-run executes the registered tool with the stored input, not the stored SQL.

Runs from a clean clone. Scales past the brief.




```
git clone --recurse-submodules https://github.com/M0ttii/log-insights-monorepo-2.git
cp .env.jury.example .env   # set password and API key
docker compose up           # download, verify, load, serve :3000
```

BEST MCP SERVER

8 tools over stdio for Claude Desktop and Claude Code, one Docker command. Zod contracts state units, limits and truncation; errors return a correction hint, never a silent zero.

HUGE DATASETS

OpenTelemetry Parquet read in place over HTTPS, never downloaded. A 60-second window:

32,605,195 rows

54 of 1,624 row groups / 69.8 MiB fetched / 1.79 s wall

Plus five Loghub extensions and a bounded OTLP replay through SereneDB's own listener. Single local observation, 18 Oct 2026; README.

Question → evidence → recommendation.

SereneDB

parses, searches, counts, forms episodes and ranks. The model never computes.

Receipts

for every number: input, SQL, scope, truncation, digest, re-run.

Roles

change the question, the evidence and the decision, not the length.

github.com/M0ttii/log-insights-monorepo-2

README: setup, tool contracts, MCP connection and limits / submission/: worked investigation, receipts, Claude answers

REPRODUCIBLE QUERY

Original SQL and a matching registry re-run

EXACT EXECUTED PHRASE-COUNT QUERY

```
SELECT count(*) AS total FROM log_events_idx WHERE
log_events_idx.message @@ ts_phrase($1) AND log_events_idx.dataset
= $2 AND log_events_idx.ts >= $3::TIMESTAMPTZ AND
log_events_idx.ts < $4::TIMESTAMPTZ
```

```
Bound parameters: (
  "failed to read message prefix",
  "bgl",
  "2005-06-01T00:00:00Z",
  "2005-07-01T00:00:00Z",
)
```

TOOL AND RESULT

search_logs

dataset: bgl / match: phrase

5,983 r_b63b6c6e.f4 exact matches

Bounded hits are returned separately.

Registry re-runs for the precursor, team-priority and OpenStack latency calls match signature, data version and result digest.

Exact SQL and parameters are read from the original receipt at build time. Data version: dv_229e422f86a9fc6e. Re-run invokes the registered tool with stored input, never stored SQL.

TOOL CATALOG

Eight tools the agent can choose from

FIND AND INSPECT

describe_corpus

Population, fields, versions, capabilities and role guidance

search_logs

Terms, phrase, fuzzy; UTC window, filters, limit / bounded hits plus exact total

get_log_context

Event neighbours, or aggregate precursors with base rate and ratio

check_claims

Value, metric and group against a fact ID / verified, mismatch, unresolved

MEASURE AND COMPARE

count_events

Grouped counts, shares, rest group; period comparison with R-06 class

event_trend

UTC bucket grid within one dataset's span; outside the span is not zero

recurring_patterns

Episodes, breadth, recurrence status, investigation order and dominance

latency_stats

OpenStack API-access percentiles, 4xx/5xx rates over a named denominator

Every output carries facts, a receipt ID, the row limit and a truncated flag. Unsupported dataset/function combinations return invalid_input with a hint. Full contracts: README, Tools.

LIMITS WE STATE ON PURPOSE

What the evidence does not establish

IN THIS CASE

Initiating node, job boundaries and why the working directory was unavailable remain open. The precursor is a temporal association, not a cause.

Episodes are not incidents. No downtime, cost or customer effect is established. Ownership is fictional.

IN THE EVALUATION

The role views on slide 5 are analyst-written from real tool calls; slide 6 shows selected real Claude answers. Neither is a blinded benchmark.

Database timings are recorded receipt round trips, not end-to-end model latency. Formal baseline and holdout runs are deferred.

See submission/minimum-scope.md and submission/worked-investigation.md. Original Loghub excerpts keep their attribution (CC BY 4.0).